

Braking System - High Level Design

Prototype Evaluation Document - Synthetic Data

Field	Value
Document ID	HLD-BRK-001
Version	1.0
System	Braking System
Classification	Synthetic / Evaluation Only

1. Purpose

This High Level Design describes the software architecture of the Braking System. The document defines software components, ports, interfaces, signals, dependencies and high-level functional flow for prototype evaluation.

2. Software Components

Component	Responsibility
BrakePedalSensor	Measures brake-pedal demand.
BrakeController	Calculates requested braking command.
BrakeActuator	Executes the braking command.

3. Interfaces

Interface	Provider	Consumer
BrakePedalInterface	BrakePedalSensor	BrakeController
BrakeCommandInterface	BrakeController	BrakeActuator

4. Port Definitions

Component	Port	Port Type	Interface
BrakePedalSensor	BrakePedalOut	P-Port	BrakePedalInterface
BrakeController	BrakePedalIn	R-Port	BrakePedalInterface
BrakeController	BrakeCommandOut	P-Port	BrakeCommandInterface
BrakeActuator	BrakeCommandIn	R-Port	BrakeCommandInterface

5. Signal Definitions

Signal	Interface	Unit
BrakePedalPosition	BrakePedalInterface	%
BrakeCommand	BrakeCommandInterface	%

6. Component Dependencies

Dependent Component	Required Component
BrakeController	BrakePedalSensor
BrakeActuator	BrakeController

7. Functional Flow

BrakePedalSensor -> BrakeController -> BrakeActuator

8. Revision Information

Current document version: 1.0. Document identifier: HLD-BRK-001.